

Effectiveness of Temporary Portable Rumble Strips in Utah

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Abstract

The abstract is a crucial component of any scientific paper, as it provides a summary of the research and its main findings. This paper provides guidelines for writing an effective scientific abstract. The first step is to identify the key elements of the research, such as the research question, methods, results, and conclusions. Next, the abstract should be written in a clear and concise manner, using simple language and avoiding technical jargon. The abstract should also be structured, with a clear introduction, methods section, results section, and conclusion. Additionally, the abstract should accurately and succinctly convey the main findings of the research, highlighting the significance and implications of the work. By following these guidelines, researchers can ensure that their abstract effectively communicates the key aspects of their research and attracts the attention of potential readers. - Written by ChatGPT

Preface

This is a template repository that I and my students can use to start projects that will implement the workflow presented in my [lab documentation](#). It also serves as an instruction manual in this workflow, a template article, and a sandbox for me to practice and learn. I encourage students to use the [Quarto Guide](#) as their primary reference.

The document in this template renders to two¹ outputs:

- A website
- An Elsevier journal article

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¹I hope to make it possible to render the article to a BYU Engineering thesis as well. Give me a bit of time.

To render this document, use the command `quarto render` in your terminal pointed at the working directory. This will create a website available locally in a `_book` folder and a PDF of the article stored in that folder.

To render your website *and* push its content to a live website, use the command `quarto publish gh-pages`. Details of this process are available on the [Quarto guide](#).

You can change the article to a different publisher by following the directions at the [Quarto Journal Templates GitHub](#) repository.

1. Introduction

The introduction of your report is not simply an “introduction”, but rather a **motivation** of why your project matters. What is the cost of not solving this problem? What have been previous attempts to solve this problem? The *why* is more important than the *what*. Why is this article worthy of archiving?

The introduction to an article is usually three or so paragraphs long. First, state the overall context of the problem, with citations to basic statistics and previous major findings. Then, Discuss the specific context of your research; why have previous methods or research not addressed your specific issue? You also need to have citations here.

In the third paragraph, you can introduce the purpose of your research and the new method or data you bring to the issue. This paragraph often begins with a statement similar to “In this paper, we present...”

An outline of the article is then usually a good idea, even though the outline might change little from article to article. A Literature Review details prior research in this area, and discusses its strengths and limitations. A Methodology section identifies the data, models, and other elements you use, while a Results section presents the outcomes of your methodology with tables and figures and a comprehensive discussion of their meanings. Discussions, and Conclusions sections may present your interpretations of your findings for future research, policy, etc.

2. Literature Review

Effectiveness of Temporary Portable Rumble Strips (TPRS)

2.1. Overview

Non-adhesive TPRS, typically made of steel and rubber, have been used since 2009 to improve work zone safety (**pss_best_2018?**). Research consistently shows that TPRS can reduce traffic speed, increase driver attentiveness, and minimize erratic driving behaviors such as lane departures (Yang et al., 2015).

2.2. Speed Reduction

TPRS installations are effective at reducing vehicle speeds in work zones. Studies report speed reductions ranging from 3.5 to 13.8 mph, (Yang et al., 2015), with greater reductions observed near the first set of strips and when drivers can see a flagger or additional barriers (wang_effects_2013?). (wang_effects_2013?) shows the effect is highly localized, and speeds tend to increase once drivers pass the TPRS unless additional measures are present.

2.3. Driver Behavior

TPRS increase driver attentiveness, often measured by the frequency of brake light activation as vehicles approach the strips. Studies show that TPRS prompt more drivers to brake and pay attention to changing road conditions, though the percentage varies from 11 percent to 55 percent based on site and conditions (Yang et al., 2015; brown_effectiveness_2022?; sun_elevated-risk_2011?; wang_effects_2013?). Commercial vehicles tend to respond more cautiously than passenger cars (wang_effects_2013?).

Concerns about drivers evading TPRS by leaving their lane have been largely unfounded. Most studies report minimal lane departures, especially when TPRS are installed across the full width of the road (sun_elevated-risk_2011?; wang_effects_2013?; brown_effectiveness_2022?). Lane departures are more likely on narrow roads or near the end of a TPRS array, but overall rates are low (sun_elevated-risk_2011?).

2.4. TPRS Displacement

A key challenge with non-adhesive TPRS is their tendency to shift or become displaced under traffic. Displacement increases with heavier vehicles, higher speeds, and higher traffic volume (sun_elevated-risk_2011?). Placement angle also matters—strips that rotate away from perpendicular to traffic tend to move more (sun_elevated-risk_2011?). Real-world studies show that while displacement is generally manageable, it can require periodic adjustment, which may increase worker exposure to traffic. The effect of strip wear and long-term durability under heavy use is not well understood and requires further research.

2.5. Spacing Specifications

There are three main groups of spacing specifications. Some states, such as Wisconsin, Florida, and Colorado, specify a single spacing independent of speed (citation?). Some states such as Maryland, Virginia, Minnesota, and Utah vary spacing from 10 to 20 ft. based on the posted speed limit (citation?). New York, Texas, and Missouri specify spacing from 10 to 35 ft. based on the posted speed limit (citation?). The manufacturer also recommends spacing of 10 to 35 ft. based on the posted speed limit (citation?). The various specifications are shown in Figure Figure 1.

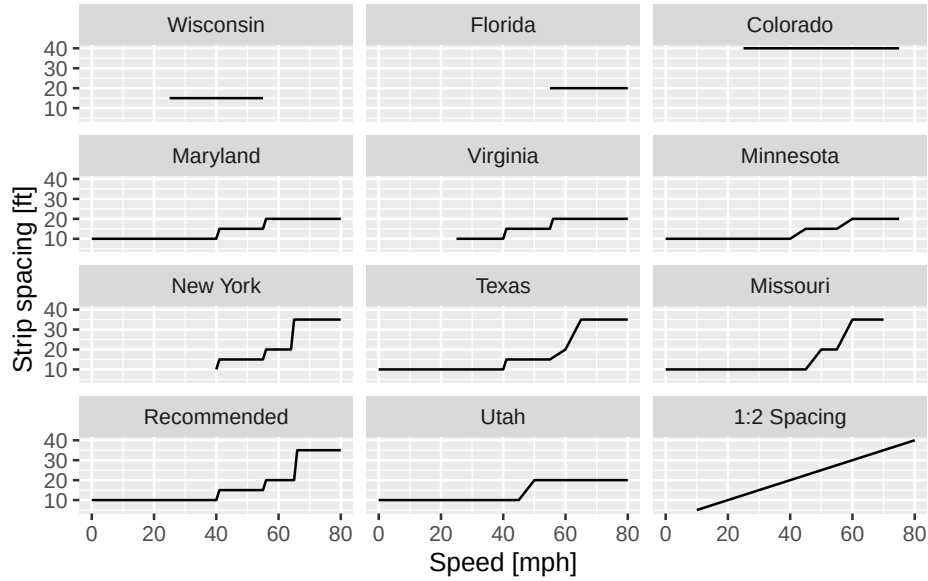


Figure 1: Various specifications for TPRS spacing.

2.6. Summary

TPRS are effective at reducing speeds, increasing driver attentiveness, and minimizing erratic driving in work zones. Displacement under traffic remains a practical concern, especially for non-adhesive products, and more research is needed on long-term performance and optimal deployment strategies.

3. Methodology

This chapter describes the methodology designed by the research team to evaluate the effectiveness, durability, and stability of TPRS systems on high-speed roadways in Utah. The experiment will observe driver behavior and TPRS displacement under traffic load. Cameras and Wavetronix units will be used to collect the desired data. The designed experiment will observe the results that occur in the absence of TPRS and in layouts that vary the spacing between the individual strips in arrays. The experiment will be designed to maximize observation time and minimize how often contractors will need to adjust the TPRS. Work sites were selected for observation based on traffic volume, availability, and location.

3.1. Site Selection

The team selected work sites available during the summer of 2025 based on their driving distance from Provo. The research team used 2023 AADT data to estimate how many vehicles typically traveled on each road segment associated with a work site being considered for observation. This estimate ensured that the

Table 1: Data Collection Sites

Site	Roadway	Location	Start	End
1	SR-12	Mile marker 6.5	2025-07-07	2025-07-11
2	US-6	20 miles W of Wellington	2025-07-15	2025-07-18
3	I-70	Clear Creek Summit	2025-07-29	2025-08-01
4	US-191	North of summit	2025-08-05	2025-08-08

roads chosen for observation had enough traffic to support a reliable statistical analysis of the potential effect spacing would have on TPRS arrays.

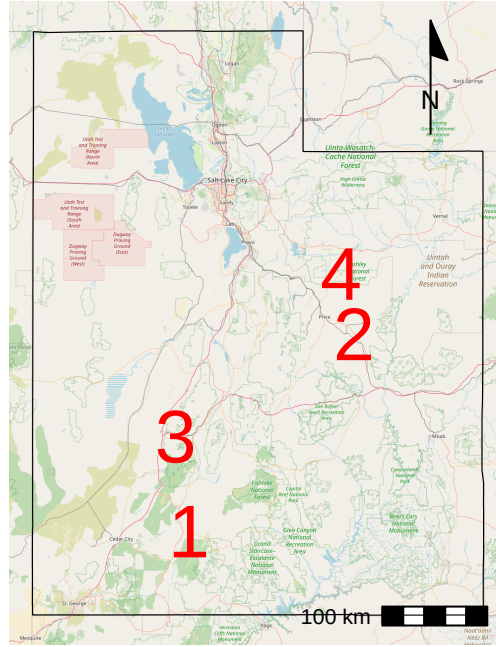


Figure 2: Map of sites

The research team used histograms of haverage hourly volumes on Utah roads to determine which ones would make reliable sites to study to ensure statistical significance. Figure 3 shows data from four counting stations near the sites selected for observation. Volume data from May 2023 to August 2023 was collected from the UDOT Traffic Data Portal, aggregated and averaged by hour, and plotted as histograms. The data shows that the sites selected for observation have a sufficient volume of traffic to support a reliable statistical analysis of the potential effect spacing would have on TPRS arrays.

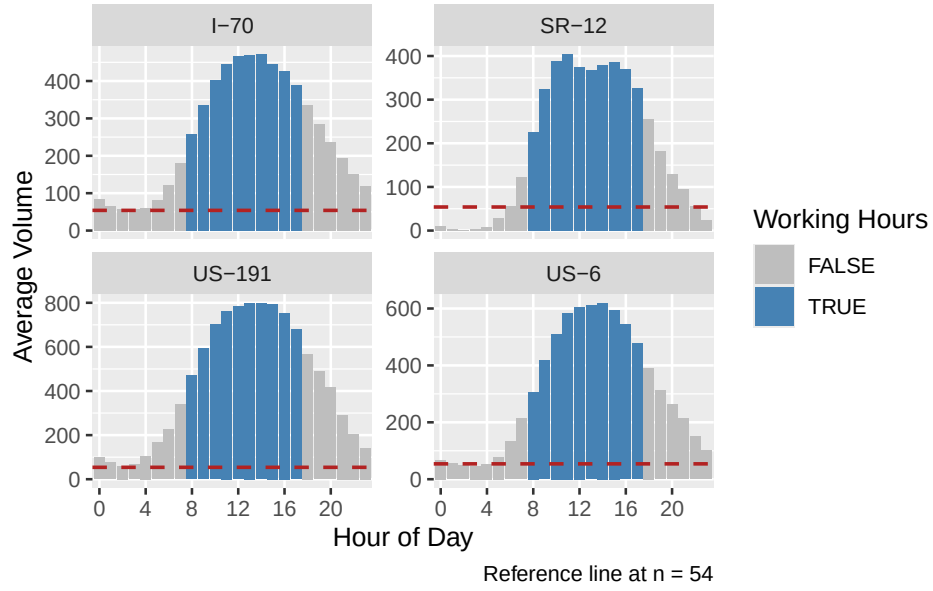


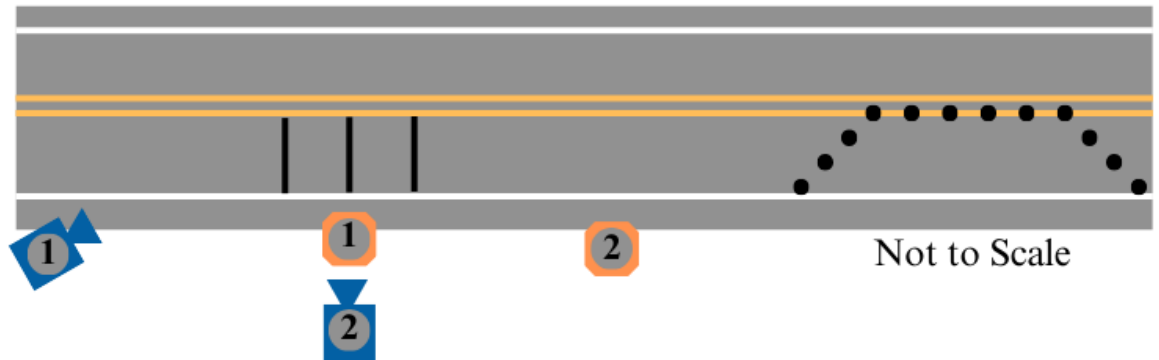
Figure 3: Histogram of average hourly volumes from counting stations near sites.

3.2. Layouts

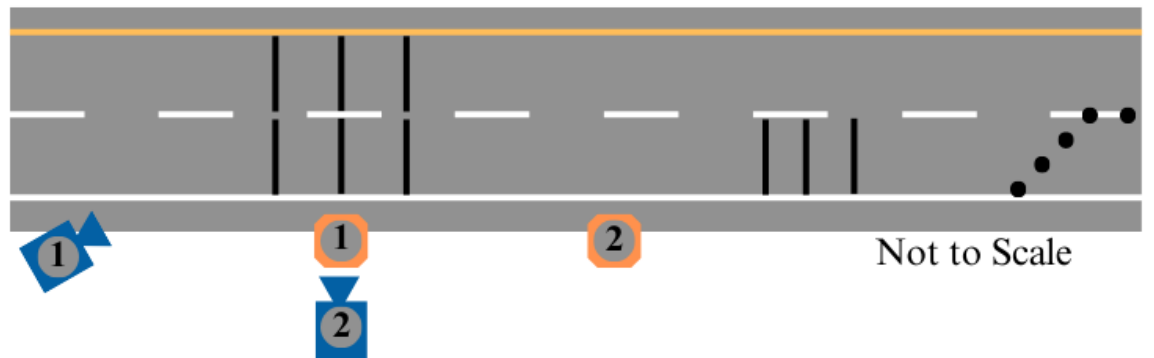
The team designed layouts to test the effectiveness of TPRS at work sites. These layouts were set up at the arrays of TPRS positioned in front of each work site preceding the tapering line of cones that marked the beginning of the work zone. Each layout consisted of two Wavetronix radar detectors and two countCAM4 cameras from StreetLogic Pro. These devices were mounted on trailers and tripods stationed on the side of the road. The layouts are depicted below, showing the set-ups designed for both single-lane and passing-lane configurations.



Lane Closure



Passing Lane Closure



The research team primarily focused on testing the space between individual TPRS to observe its effect on driver speed and behavior. They also examined whether the strips could remain in place without requiring frequent readjustments. The team developed a spacing system for contractors to use during observation days, tailored to the posted speed limit at each site. The different spacing specifications, shown in Figure 4 were derived from current UDOT standards for TPRS spacing and an unofficial rule of thumb from Plastic Safety Systems (PSS) for their RoadQuake TPRS system.

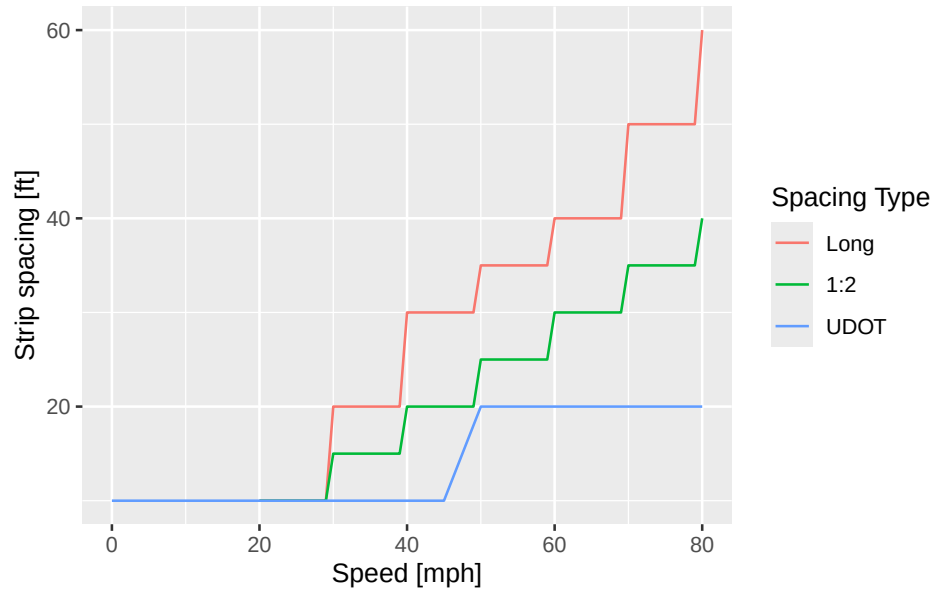


Figure 4: Spacing specifications used in testing.

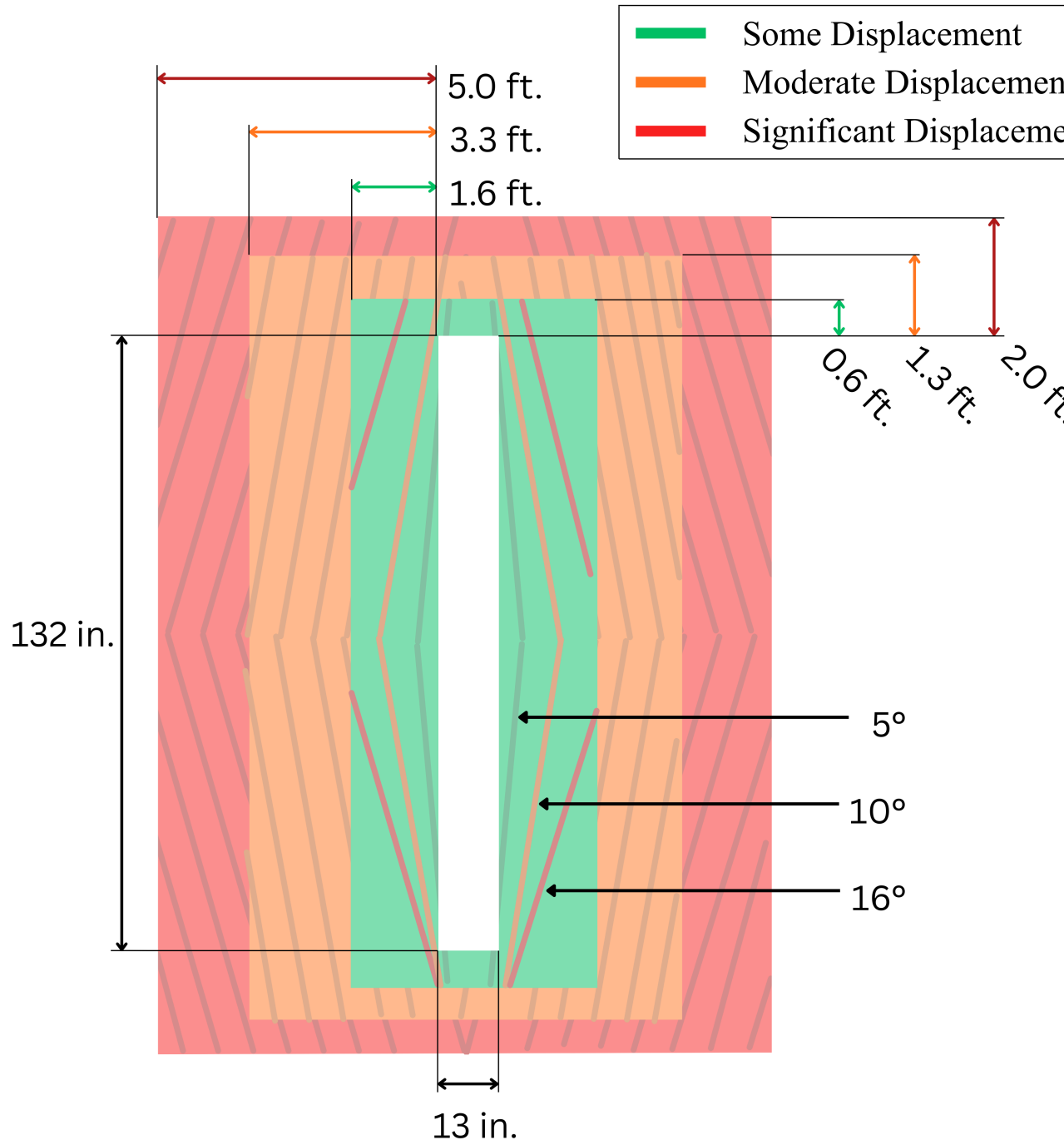
Table 2: Separation [ft] Between Wavetronix Units

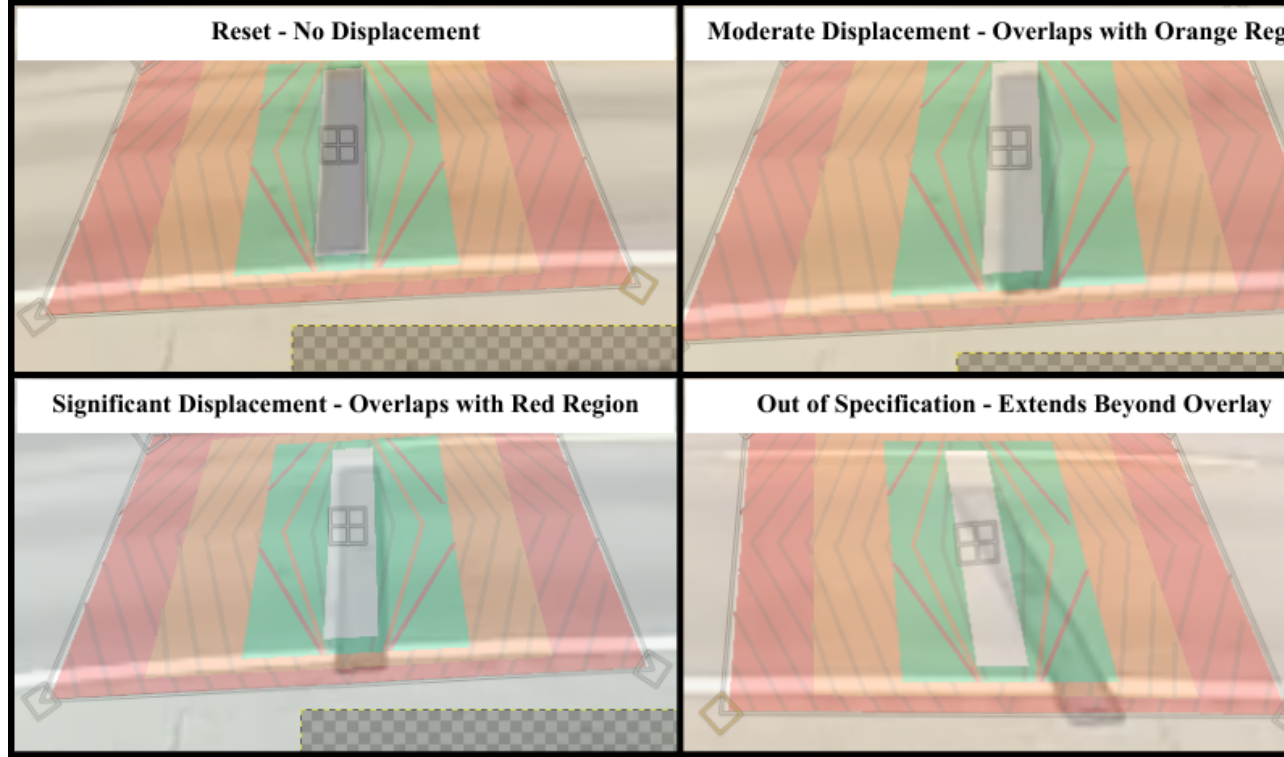
Site	UDOT	NO TPRS	1:2	LONG
SR-12	762/875	851	878	898
US-6	516	516	516	516
I-70	800	804	782	765
US-191	782	748	781	759

3.3. Driver Behavior - Change in Speed, Braking, and Lane Departures

The research team observed driver behavior by analyzing the data collected from the Wavetronix radar detectors and StreetLogic Pro cameras. The Wavetronix units provided data on vehicle speed and braking patterns, while the cameras captured footage to identify lane departures and other driving behaviors. The team compared these behaviors across different spacing configurations to assess the effectiveness of TPRS in influencing driver behavior in work zones.

3.4. TPRS Displacement and Worker Exposure





4. Results and Analysis

This section presents the results of the methodology in Chapter 3.

4.1. Speeds

Figure 5 shows the change in speed between units, 95% confidence bound of different between 85th percentile speeds. The 95% confidence bound is the range of speeds that are likely to occur if the null hypothesis is true. The confidence bounds are calculated using the bootstrap method. The 95% confidence bound is calculated by taking the upper and lower bounds of the 95% confidence interval, and then taking the difference between them. This is a useful metric for understanding the effect of rumble strips on traffic flow.

4.2. Braking Behavior

4.3. Avoidance Behavior

4.3.1. Strip Displacement

Figure 7 shows the impact momentum for each transition, colored by site. The impact momentum is the change in speed for a vehicle in a given transition. The change in speed is calculated by subtracting the speed at the beginning of the

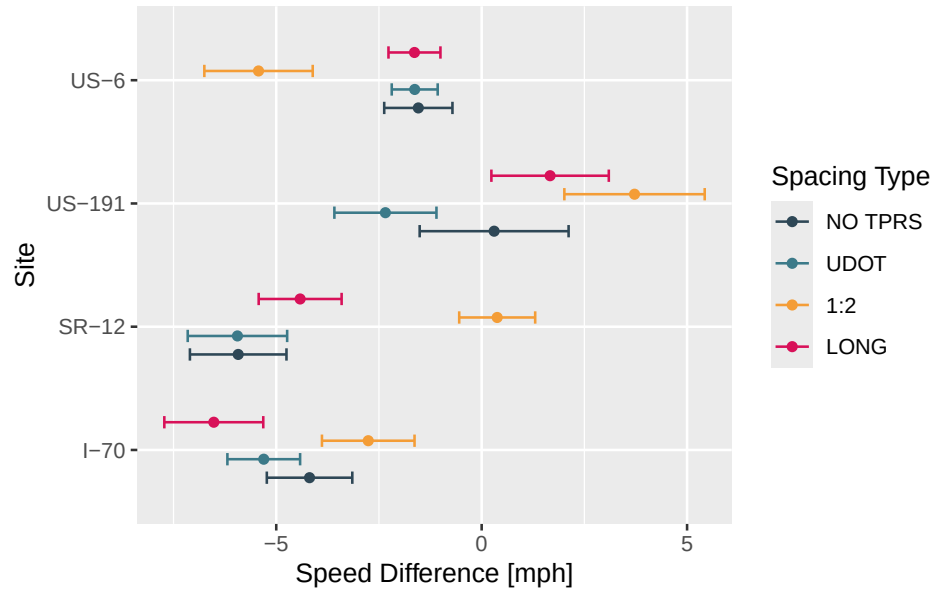


Figure 5: Change in speed between units, 95% confidence bound of different between 85th percentile speeds.

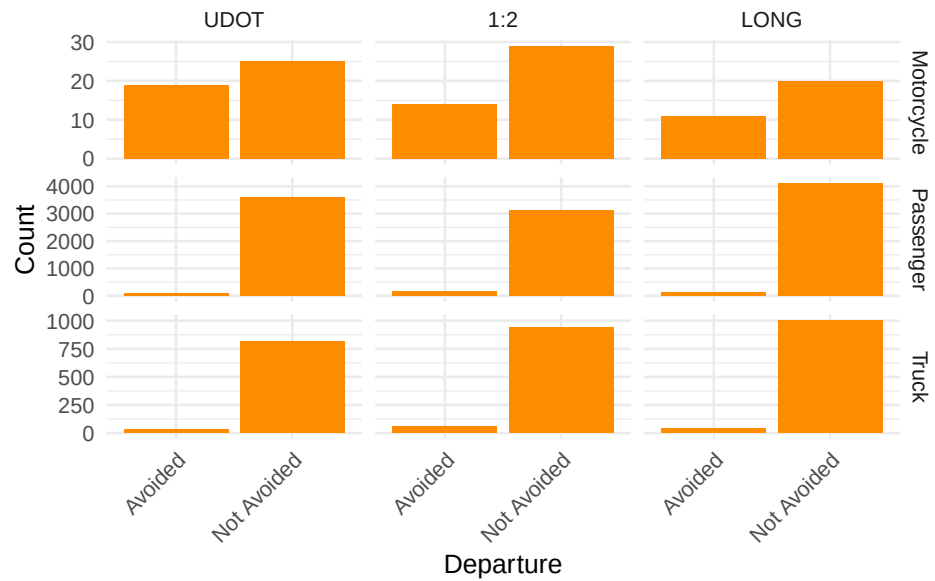


Figure 6: Avoidance behavior by spacing of TPRS and vehicle type.

Table 3: Binomial Models of TPRS Avoidance

	Pooled	SR-12	US-6	I-70	US-191
(Intercept)	-7.852*** (0.382)	-5.622*** (0.413)	-7.577*** (1.012)	-21.499 (754.412)	-6.343*** (1.009)
Trucks	0.166 (0.117)	0.306 (0.204)	0.536 (0.364)	-0.669+ (0.388)	0.121 (0.172)
Motorcycles	2.213*** (0.210)	2.192*** (0.282)	2.874*** (0.777)	3.836*** (0.948)	2.060*** (0.354)
UDOT Spacing	3.189*** (0.367)	2.868*** (0.433)	1.875+ (1.063)	16.662 (754.413)	4.313*** (1.013)
1:2 Spacing	3.538*** (0.363)	3.092*** (0.423)	3.145** (1.043)	17.971 (754.412)	4.312*** (1.012)
Long Spacing	3.292*** (0.365)	3.165*** (0.424)	2.723** (1.030)	15.184 (754.413)	4.071*** (1.016)
SR-12	1.947*** (0.153)				
US-191	2.352*** (0.165)				
US-6	-0.332 (0.213)				
Num.Obs.	19016	4963	6324	5933	1796
AIC	4056.3	2018.0	454.8	530.1	1022.9
BIC	4126.9	2057.1	495.3	570.2	1055.9
Log.Lik.	-2019.131	-1003.004	-221.398	-259.041	-505.462
RMSE	0.16	0.23	0.08	0.09	0.28

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

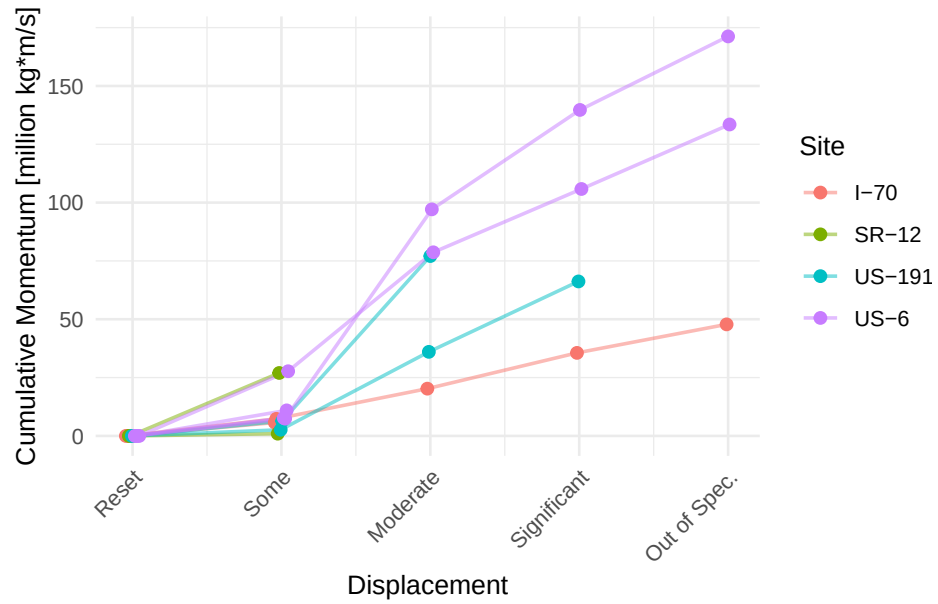


Figure 7: Impact momentum for each transition, colored by site.

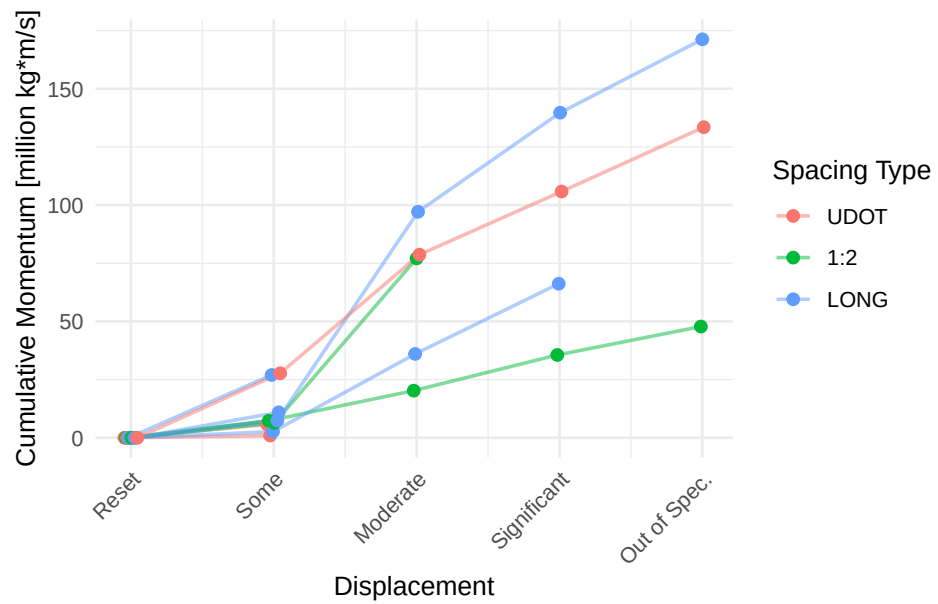


Figure 8: Impact momentum for each transition, colored by spacing.

transition from the speed at the end of the transition. This is a useful metric for understanding the effect of rumble strips on traffic flow.

4.3.2. Worker Exposure

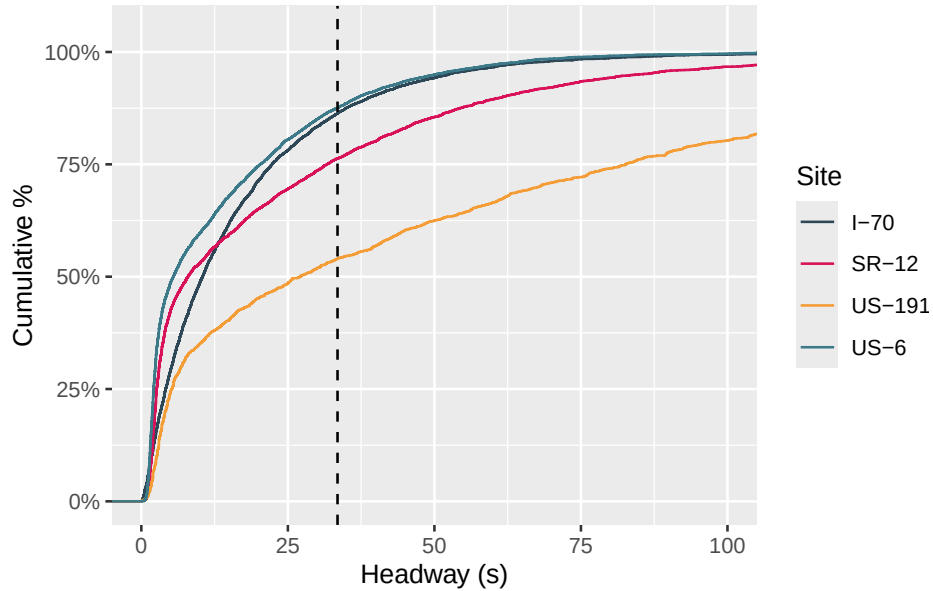


Figure 9: CDF of headways for each site.

5. Conclusions

This section need not be overly long. You should address any limitations of your results, such as dependence on underlying assumptions or geographic scope. You should also provide a map for future research.

Finally, you should underline the contributions of this work and any practical relevance.

References

Yang, H., Ozbay, K., & Bartın, B. (2015). Effectiveness of temporary rumble strips in alerting motorists in short-term surveying work zones. *Journal of Transportation Engineering*, 141(10), 05015003. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000789](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000789)